



## 5.6 Further calculus applications to modelling (HL)

Calculus · IB Diploma · Concept guide and original exam-style practice

### Understand the concept

Related rates problems use the chain rule to connect the rates of change of two linked quantities, e.g.  $dV/dt = (dV/dr) \times (dr/dt)$ . HL calculus also applies integration to find the area between two curves,  $\text{Area} = \int [a, b] (f(x) - g(x)) dx$  where  $f(x) \geq g(x)$  on  $[a, b]$ , and uses the second derivative to analyse concavity and locate points of inflection (where  $f''(x) = 0$  with a sign change).

### Worked example: follow the reasoning

A spherical balloon is inflated so its volume increases at a constant rate of  $50 \text{ cm}^3/\text{s}$ . Find the rate at which the radius is increasing when the radius is 5 cm, given  $V = (4/3)\pi r^3$ .

Solution:  $dV/dr = 4\pi r^2$ . By the chain rule,  $dV/dt = (dV/dr) \times (dr/dt)$ , so  $dr/dt = (dV/dt)/(dV/dr) = 50/(4\pi \times 25) = 50/(100\pi) = 0.5/\pi \approx 0.159 \text{ cm/s}$ .

### Quick check · multiple choice

Related rates problems primarily rely on which calculus rule to connect two rates of change?

- A Product rule
- B Chain rule
- C Quotient rule
- D Power rule only

Write your choice: \_\_\_\_\_

### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The area  $A$  of a circular oil spill increases as its radius  $r$  increases, with  $dr/dt = 0.4$  [8]  
 m/min (constant), where  $A = \pi r^2$ . (a) Find  $dA/dr$  in terms of  $r$ . (b) Use the chain rule  
 to find  $dA/dt$  in terms of  $r$ . (c) Find the rate at which the area is increasing when  
 the radius is 12 m, to 1 decimal place. (d) Find the area enclosed between the  
 curves  $y = x^2$  and  $y = 2x - x^2 + 4$  for  $0 \leq x \leq 2$ , using integration.

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Final answer: \_\_\_\_\_



## Answers and explanation

### Question 1 [8]

(a)  $dA/dr = 2\pi r$ . (b)  $dA/dt = (dA/dr) \times (dr/dt) = 2\pi r \times 0.4 = 0.8\pi r$ . (c) At  $r=12$ :  
 $dA/dt = 0.8\pi(12) = 9.6\pi \approx 30.2 \text{ m}^2/\text{min}$ . (d) Setting  $x^2 = 2x - x^2 + 4$  gives  $2x^2 - 2x - 4 = 0 \rightarrow x^2 - x - 2 = 0 \rightarrow (x-2)(x+1) = 0$ , so on  $[0, 2]$  the curve  $y = 2x - x^2 + 4$  lies above  $y = x^2$  (checked at  $x=0$ :  $4 > 0$ ).  
 $\text{Area} = \int_{[0,2]} (2x - x^2 + 4) dx = [x^2 - (2/3)x^3 + 4x]$  from 0 to 2 =  $12 - 16/3 = 20/3 \approx 6.67$ .

### Quick check answer

B. The chain rule links  $dA/dt$  to  $dA/dr$  and  $dr/dt$  (or similar pairs of connected rates).

### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.